Software Testing

Black Box Testing

Black box testing is a testing method where the internal mechanism and code are not examined. The tester provides selected inputs and checks the resulting outputs.

Advantages:

(i) Robust with respect to implementation changes.

(ii) Allows independent testers

(iii) Testers need not be familiar with the code.

(iv) Can be less time consuming

Disadvantages

(i) may miss implementation bugs

(ii) Does not directly guide internal optimization or alternate algorithms.

White Box Testing

White box testing is a testing method that takes the internal mechanisms and code into account. It tests internal logic, conditions, loops and branches.

Advantages

(i) Allows through testing of the software's internal logic.

(ii) Helps find hidden bugs and errors

(iii) Yields useful test cases

(iv) Enables code optimization and experiments with alternate algorithms.

Disadvantages

1. The number of path grows rapidly; loops can make it unbounded.
2. Requires access to the code.
3. Requires a skilled tester with detailed knowledge of the internal system.
4. More time consuming
5. Exhaustive path testing is often impractical

Functional Testing

Functional Testing checks whether the software functions/features work according to the Functional Requirement Specification (FRS).

It gives inputs and checks the outputs.

Ex: Check a coupon, is that valid or not.

Non-Functional Testing / Technical Testing

Non-functional Testing checks aspects such as performance, usability, reliability and security, rather than only checking whether a feature produces the correct result.

Ex:

একটি checkout system.

Functional: checkout -এ Total কি correct?

Non-Functional: 500 simultaneous users থাকলেও checkout কি 2 seconds এর মধ্যে response দেয়?

Unit Testing

Unit Testing tests individual software or hardware units, or groups of related units. It verifies that the code does what it is intended to do at a very low structural level.

Ex:
check one function.

calculateTotal (100, 2)

Expected outcome:

200

~~Usally~~ Usually proggrammer participate in this test, White box knowledge used here.

Integration Testing

Integration Testing combines software components, hard ware components, or both to evaluate their interactions.

Ex:

দুটা module আলাদাভাবে কাজ করতে পারে।
কিন্তু একসাথে করলে data mismatch হবে।
Integration Testing এই ধরনের সমস্যা ধরতে পারে

System Testing

System Testing tests a complete, integrated system to evaluate whether it complies with its specified requirements. It checks both functional and non-functional requirements.

Ex:

Unit Testing – এ আমরা ছোট ছোট একটা function চেক্ষি।
Integrated system এ কয়েকটা components চেক্ষি।
System Testing – এ পুরো system চেক্ষি।

One ecommerce website:

Login → search product → Add to Cart → payment → Order confirmation.

Load Testing

Load Testing is performance testing under real-life / expected load conditions. It uses concurrent users and transactions over a set duration and measures response time and

operating capacity.

Stress Testing-

Stress Testing- evaluates the system beyond its specified limits. The pressure is increased until performance degrades or failure ~~pain~~ occurs.

Ex:
Website এর expected load = 500 users.

Stress Testing:

500 → 1000 → 2000 → 3000 → ....

এভাবে load বাড়াতে থাকবো।

যখন system slow হয়ে যায় বা crash করে, তখন বোঝা যায় system এর Breaking point কোথায়।

Security Testing

Security Testing is done to find vulnerabilities and check whether data and resources are protected from possible intruders.

1. Vulnerability scanning :
   Known vulnerabilities খুঁজে বের করার জন্য automated scanning করা।

2. Ethical Hacking :
   Authorized ভাবে security weakness খুঁজে বের করার চেষ্টা করা।

3. Penetration Testing :
   controlled এবং simulated attack চালিয়ে exploitable vulnerability খুঁজে বের করা।

Regression Testing

Regression Testing is a selective retesting to verify that modifications have not caused unintended effects and that system still meets its requirements.

Ex:

Shopping Website -এ নতুন coupon feature যোগ করা হলো।

এখন শুধু coupon test করলেই হবে না।

আরও:

- Total Calculation
- Payment
- Refund

এগুলোও আবার Test করতে হবে।

Beta Testing

Beta Testing is when an organization gives an advance partial or full version of software to potential users, often for free. The user use it and report errors they find.

Ex: New shopping app

100 জন user কে beta version দেওয়া হলো।

Acceptance Testing

Acceptance Testing is formal testing performed to determine whether the system satisfies its acceptance criteria. It allows the customer to decide whether to accept the system.

Software customer-এর জন্য agreed requirement acceptance criteria পূরণ করছে কি না পরীক্ষা করাই Acceptance Testing।

মূল উদ্দেশ্য:
Customer কি software-টি accept করবে?

এটি মূলত Black Box Activity.

Testing Hierarchy

Unit Testing → Integration Testing → System Testing → Acceptance Testing

Static and Dynamic Testing

Static Testing

Static Testing examines something without running it. It reviews ~~that~~ the work rather than executing the software.

Software run না করেই code / document / design চেক্ করে problem খুঁজে বের করাকে static Testing বলে।

Ex:

ধরো code-এ আছে:

```
if total > 1000
    give discount
```

কিন্তু requirement বলছে:

```
total >= 1000
```

Dynamic Testing

Dynamic Testing works with running software. It provides inputs and checks the expected outputs. Examples include unit, integration, system and acceptance Testing.

Ex:

Discount system - ৭

total = 1000

কিছু program run করলাম।

Expected:

Discount should be given

কিন্তু discount না দিলে → bug পাওয়া গেল।

এটা Dynamic Testing.

Code smell

Code smell means code components that violate sound development principles and reduce code quality. They negatively affect readability and maintainability.

Code Review

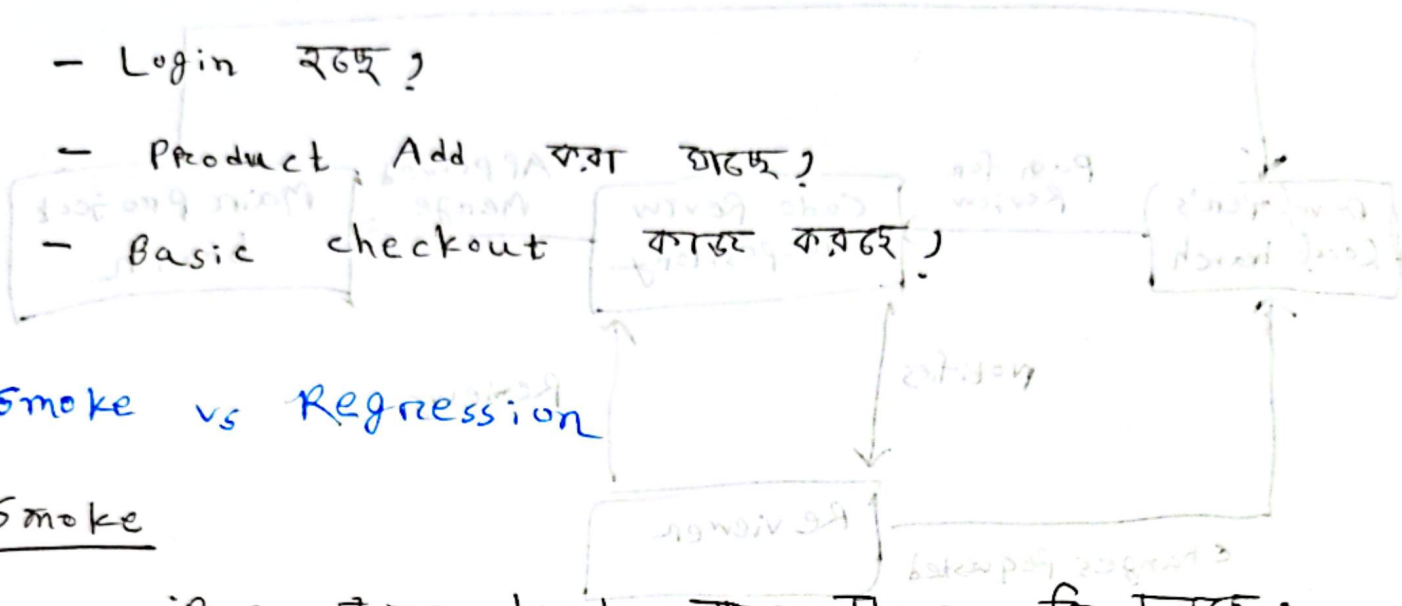
Code Review is when another developer examines a proposed code change before it becomes part of the main project codebase.

Smoke Testing

Smoke Testing is a small set of tests used to check the basic stability and major functionality of a software system under normal conditions. It is often taken from the regression test suite.

পুরো software এর সব feature একবারই test না করে ~~সবার~~ আগে কয়েকটা important / basic feature check করবে।

- Login হচ্ছে?
- Product Add করা যাচ্ছে?
- Basic checkout কাজ করছে?



Smoke vs Regression

Smoke

Basic Important ~~Thing~~ Things কি চলছে?

Regression

নতুন change এর কারণে ~~সব~~ আগের functionality নষ্ট হয়েছে কি?

Code Review Efforts

Developers = 50000

প্রতি Developer = 6 hours / week

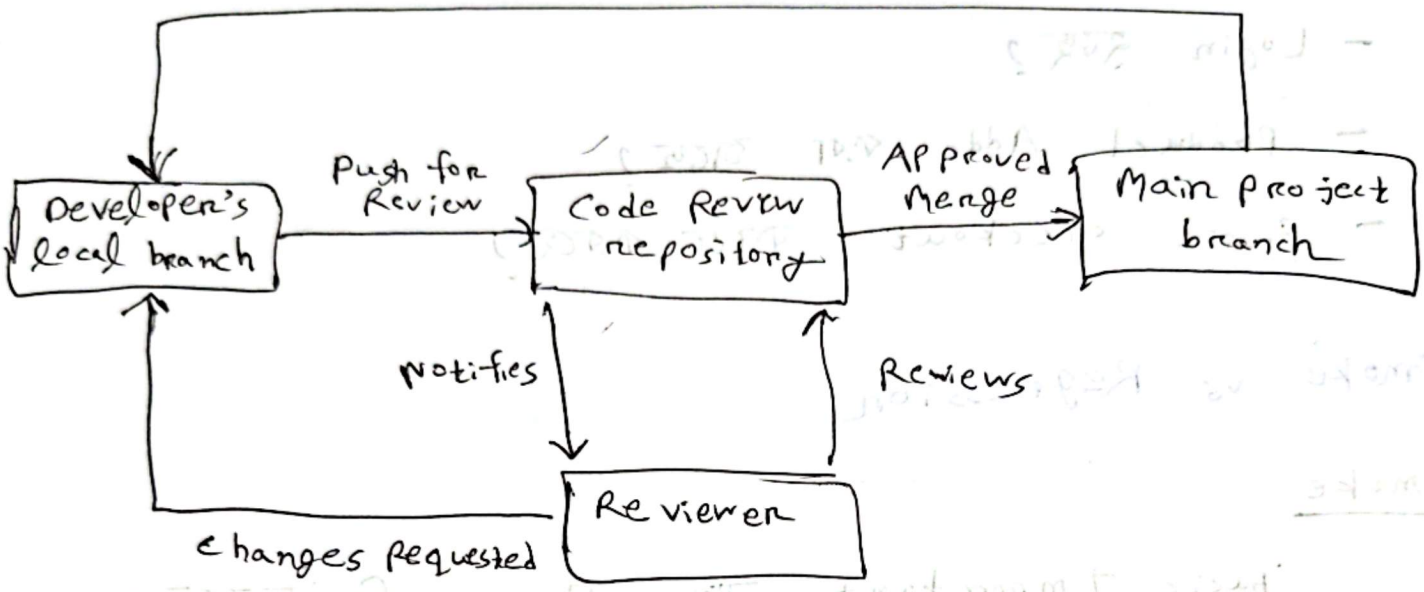
প্রতি hour = $50

calculation:

50000 × 6 × 50

= $15,000,000 / week

Code Review Workflow



Code Review Tools / Code Review Issues

↳ এগুলো ওইটাতে exam ৭ আসবে না।